

TRIG/PRECALC
REVIEW – TRIG GRAPHS & UNIT CIRCLE

NAME Key

PART I: Identifying Transformations

Determine the following transformations given the function:

$y = 5 - 2\sin(4x)$

$y = -3\tan(4x - \pi) + 1$

3. $y = \sec\left(\frac{1}{2}x + 3\pi\right)$

1.
amplitude: 2

2.
a-value: -3

amplitude: 1

period: $\frac{\pi}{2}$

period: $\frac{\pi}{4}$

period: 4π

horiz shift: None

horiz shift: Right $\frac{\pi}{4}$
(where did the inflection point go)

horiz shift: left 6π

vert. shift: up 5

vert. shift: up 1

vert. shift: None

reflection? over
~~x-axis~~
the midline

reflection? over ~~x-axis~~
the midline

reflection? None

Write the function given the following transformations:

4. cosine function

5. cotangent function

6. cosecant function

amplitude: 5

period: 8

vertically stretch by a factor of 8

shift left $\frac{2\pi}{3}$

shift right 1

vert. shift: up 2

period: 2π

reflected over ~~x-axis~~ midline

period: π

$y = 5\cos\left(x + \frac{2\pi}{3}\right)$

$y = -\cot\left(\frac{\pi}{8}x - \frac{\pi}{8}\right)$

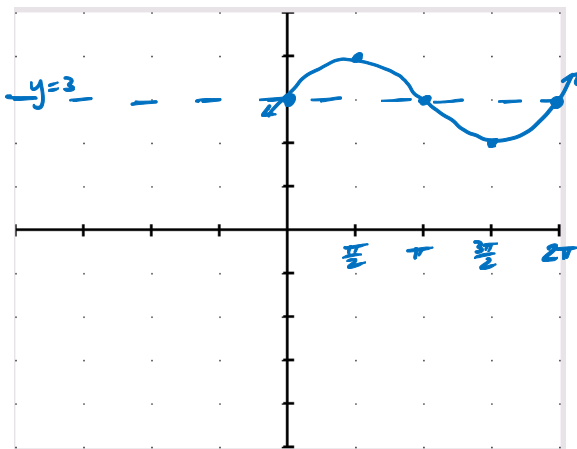
$y = 8\csc(2x) + 2$

PART II: Graphing Trig Functions

Graph the following, LABEL THE X-AXIS!!! Draw asymptotes where necessary!

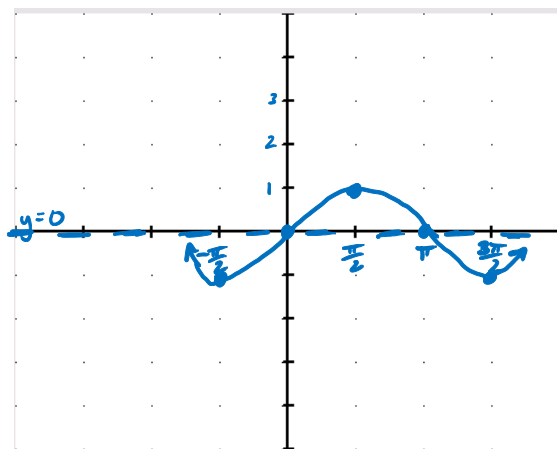
$y = \sin(x) + 3$

7.

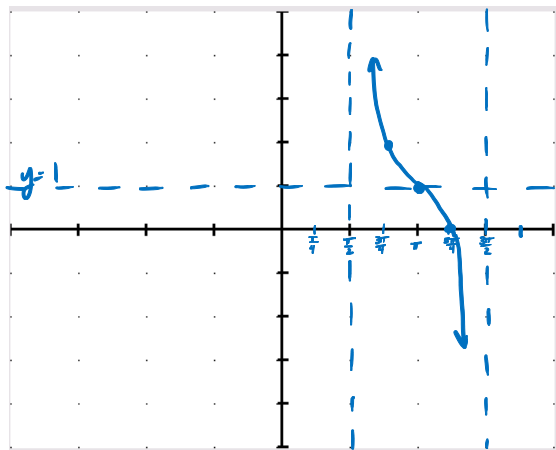


$y = -\cos\left(x + \frac{\pi}{2}\right)$

8.

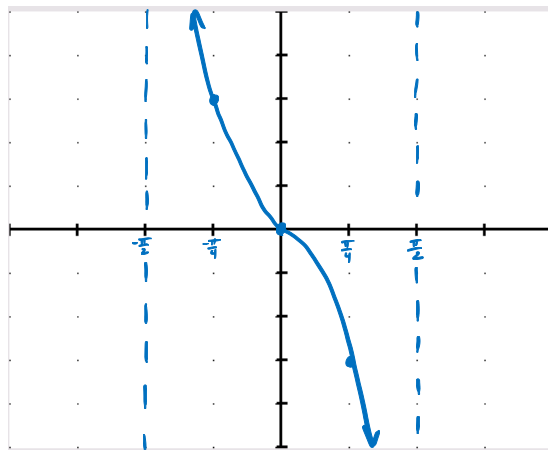


9. $y = \cot(x - \frac{\pi}{2}) + 1$



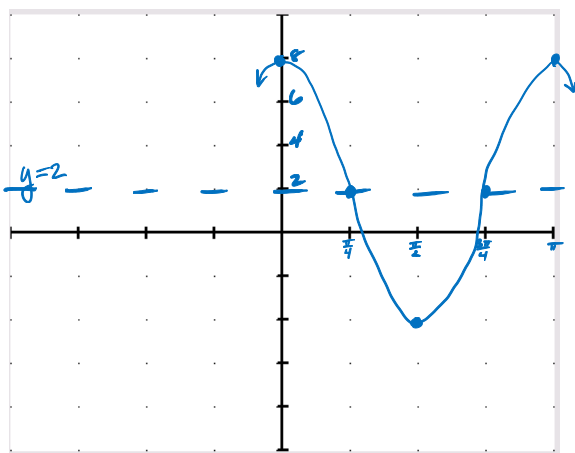
$y = -3 \tan(x)$

10.



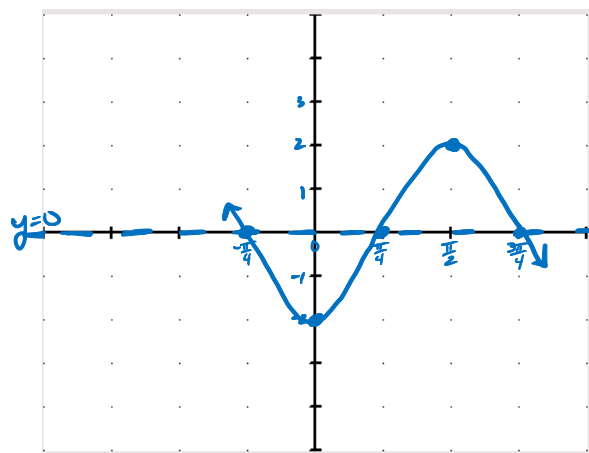
$y = 6 \cos(2x) + 2$

11.

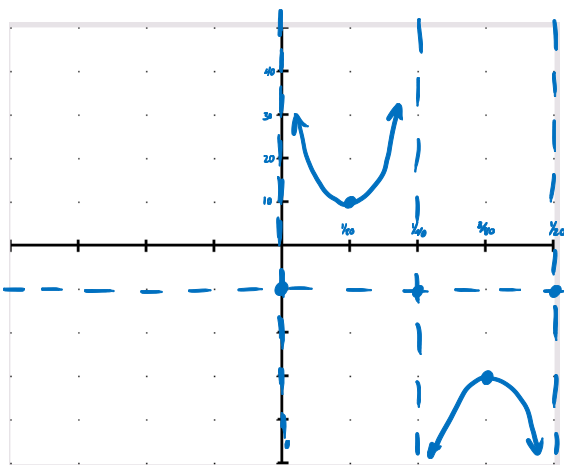


$y = -2 \sin(2x + \frac{\pi}{2})$

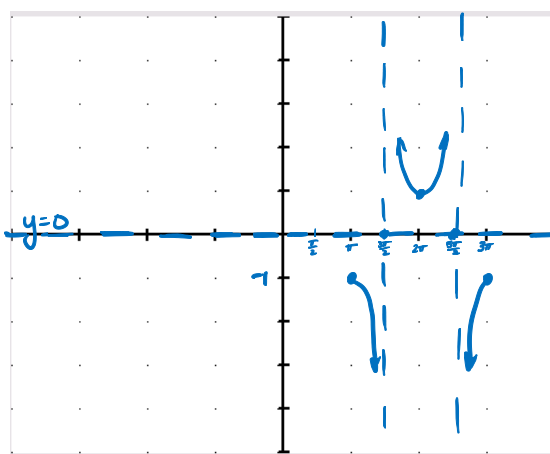
12.



13. $y = 20 \csc(40\pi x) - 10$

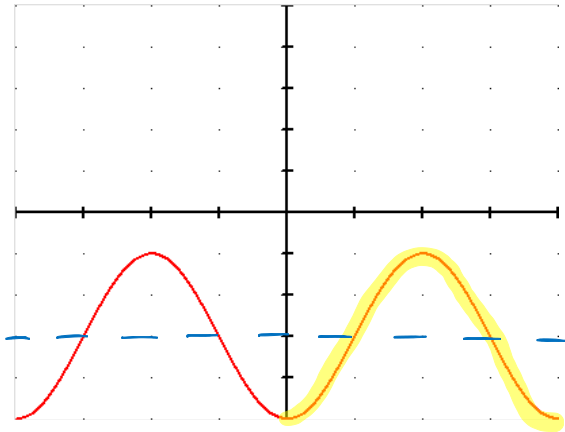


14. $y = -\sec(x - \pi)$

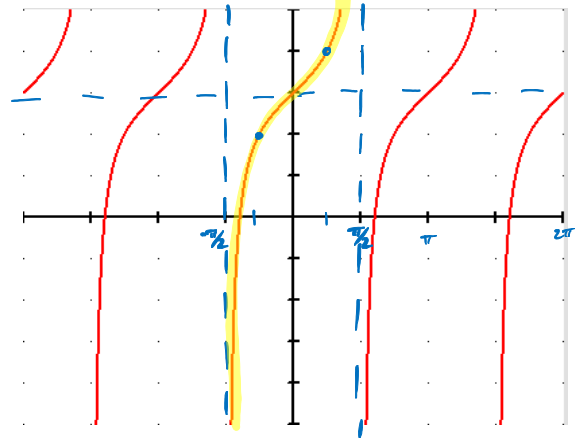


PART III: Writing Trig Equations On # 15-20 Answers will vary based on chosen LB. The equations below match the highlighted period
 Write an equation for each of the following graphs. Note that the x-axis shown on each graph is $[-2\pi, 2\pi]$

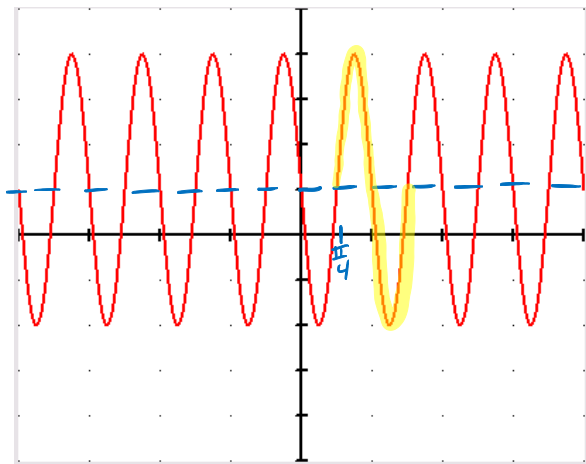
15. $y = -2\cos x - 3$



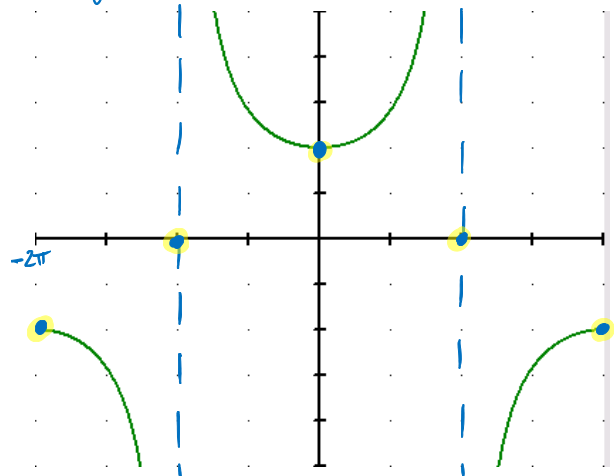
16. $y = \tan x + 3$ or $y = -\cot(x + \frac{\pi}{2}) + 3$



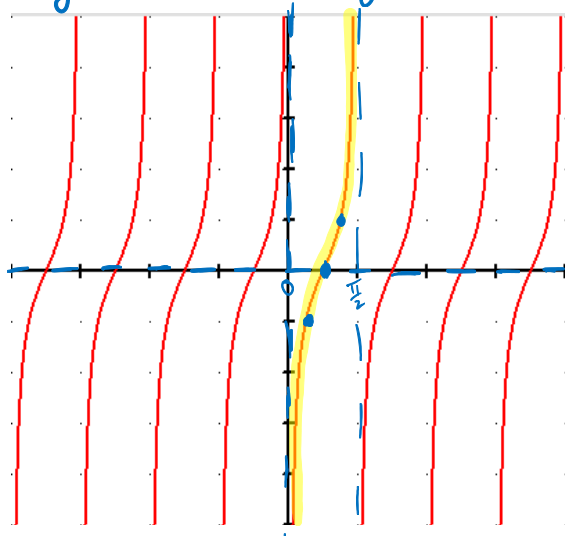
17. $y = 3\sin(4x - \pi) + 1$



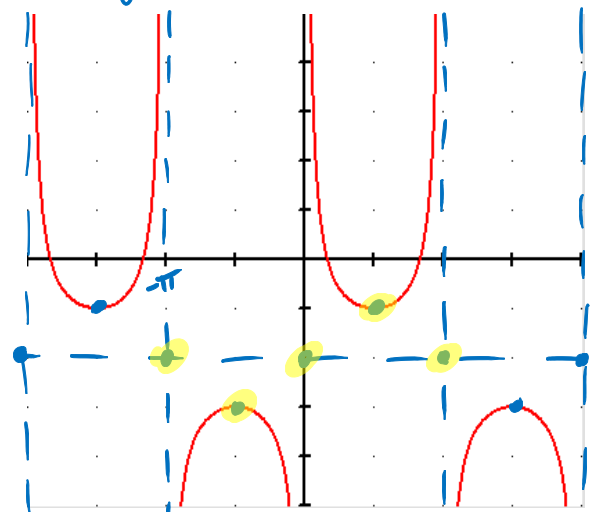
18. $y = -2\sec(\frac{1}{2}x + \pi)$



19. $y = \tan(2x + \frac{\pi}{2})$ or $y = -\cot(2x)$



20. $y = -\csc(x + \pi) - 2$



PART IV: Applications

21. The average monthly temperatures for the city of Chicago, Illinois are represented by the model (function) given below, where $x=1$ represents the month of January

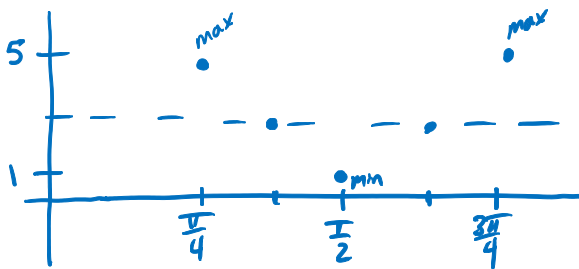
$$y = -26 \cos \frac{\pi}{6}(x-1) + 58$$

According to this model, what is the average monthly temperature in May?

* no calculator allowed

when $t=5$, $y=71^\circ$

22. Write an equation of a cosine or sine function given that there is a maximum point at $\left(\frac{\pi}{4}, 5\right)$ and a minimum point at $\left(\frac{\pi}{2}, 1\right)$.



$$y = 2 \cos(4x - \pi) + 3$$

$$y = -2 \sin(4x - \frac{3\pi}{2}) + 3$$

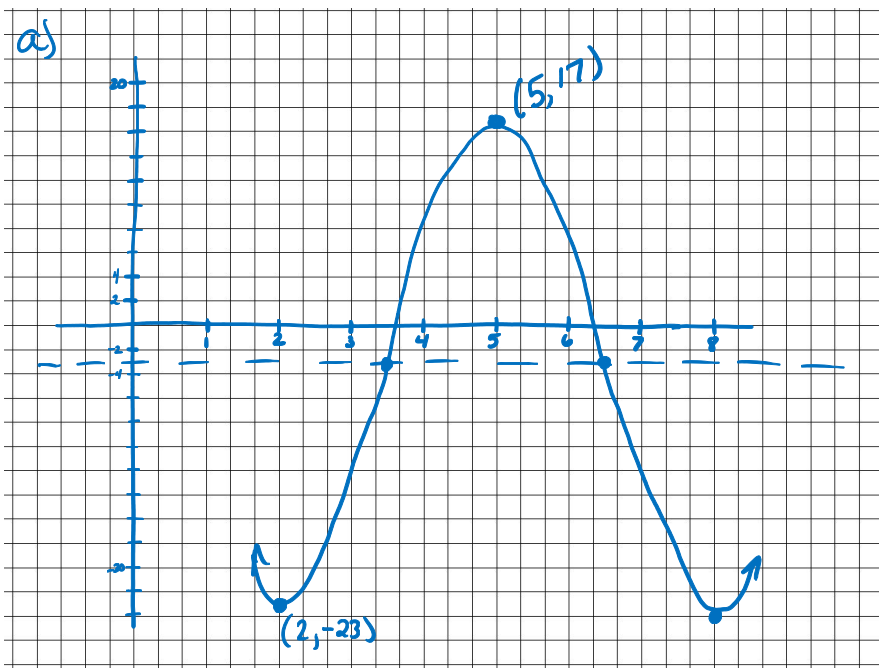
23. As you ride the Ferris wheel, your distance from the ground varies sinusoidally with time. You are in the last seat filled and the Ferris wheel starts immediately. Let x be the number of seconds that have elapsed since you got on. You find that it takes you 6 seconds to reach the top, 84 feet above the ground. The diameter of the wheel is 80 ft (you get on the ferris wheel 4 ft above the ground) and it rotates at a constant rate. Write a cosine function that models this information.

$$y = -40 \cos\left(\frac{\pi}{6}x\right) + 44$$

24. Tarzan is swinging back and forth on his grapevine. As he swings, he goes back and forth across the riverbank, going alternately over land and water. Jane decides to model mathematically his motion and starts her stopwatch. Let t be the number of seconds the stopwatch reads and let y be the number of meters Tarzan is from the riverbank. Assume that y varies sinusoidally with t , and that y is positive when Tarzan is over water and negative when he is over land. Jane finds that when $t=2$, Tarzan is at one end of his swing, where $y=-23$. She finds when $t=5$ he reaches the other end of his swing and $y=17$.

- a) Sketch a graph. b) Write an equation expressing Tarzan's distance from the riverbank in terms of t .
c) Predict y when: i) $t=2.8$ and ii) $t=15$
d) Where was Tarzan when Jane started the stopwatch?

See next page



b) $y = -20 \cos\left(\frac{\pi}{3}t - \frac{2\pi}{3}\right) - 3$

c i) * use a calculator

$$y(2.8) = -20 \cos\left(\frac{4\pi}{15}\right) - 3 \approx -16.38$$

At 2.8 seconds Tarzan is about 16.38 ft over land

c ii) * No calculator allowed

$$y(15) = -20 \cos\left(\frac{13\pi}{3}\right) - 3 = -13$$

At 15 seconds Tarzan is exactly 13 ft over land

d) * No calculator allowed

when $t=0$ $y(0)=7$

When Jane started the stopwatch, Tarzan was 7 feet over the water.